BRAC UNIVERSITY

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CSE461

Introduction to Robotics Lab

Real-World Implication

Project: Autonomous Parking Guide Robot with Smart Slot Allocation

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Introduction

Parking problems are often encountered by many drivers, especially in a busy parking facility. A driver who does not know where to park has to slowly drive around in different parking lanes, check multiple spots and park at the same one as someone else. This needless searching costs time, fuel, emissions and aggravates drivers. It might also pose safety issues from sudden stops and unexpected vehicle movements in the vicinity of pedestrians. Traditional, low cost parking systems typically use red and green lights as a rudimentary representation of free and occupied parking. These systems are not capable of authorizing vehicles, booking parking spaces ahead of time or directing the driver directly to their destination, however. The proposed Autonomous Parking Guide Robot with Smart Slot Allocation is an automatic prototype, which combines these functions into a single system. The system is based on the condition of three model parking spaces monitored by sensors and controlled by an Arduino Uno. An RFID reader determines if a vehicle coming into the lot is authorized to enter. Upon receiving approval of the card, the controller picks a free parking space, locks it for that vehicle, shows the parking space number, activates a servo-controlled entrance barrier and informs the guidance robot with the destination information. The robot then proceeds along a black line, makes turns as per the predetermined slot position and stops safely if there is an obstacle in the way using the ultrasonic sensor. This system is different from a simple available/unavailable parking indicator, as it contains information about three different states of the slots: Free: The sensor detects that no vehicle is present and can be used for an assignment.

- Reserved: This slot is already booked by an authorised vehicle and is not open for other users.
- Occupied: When a car is physically detected inside the car park slot.

This state-based approach provides for extra functions that include reservation timeout, full-parking detection, and reassignment when guided to a selected slot becomes full earlier than guidance begins. Keeping the project as a small prototype and having only fixed routes and parking places, the full system is planned to be developed, integrated and tested within a span of say about 1 month. There are two main aims of the system: Detect and classify: Ensure that the sensor reading is correct before switching between the Free, Reserved and Occupied status of the slot, to avoid false decision making as a result of temporary sensor noise. Authorize and allocate: Accept valid RFID cards, choose the closest available empty room, avoid re-using slots, indicate the assigned room, control the entrance barrier. Guide and avoid; move the selected slot data to line-following robot, run the correct preprogrammed path and end the robot when it senses an obstacle with the ultrasonic sensor.

Manage exceptions: Deny entry of unauthorized cards, do not allow entry if all the slots are full, cancel the reservation, and halt automatic operation if the sensor or communication data is not reliable.

Purpose

This project's primary objective is to show one entire autonomous parking assistance workflow—not a set of distinct hardware functions. The aim of the system is to show how all of the components of the system (sensing, access control, parking allocation, wireless communication, barrier control, mobile robot navigation, and display of user information) can be combined into an overall system. The operation is successful when the user presents an RFID card, and successful when the guidance robot arrives at the designated parking spot with the controller keeping track of the status of each parking spot. The design approach is also to have a realistic solution, which is developed in limited time by a team of the university's project. The system has no complicated technologies such as camera vision, image processing, free autonomous navigation, digital map or even vehicle charging, control of an actual car. Rather, the robot is controlled by a known set of "junction points" and a line-following technique. The team can focus on dependable integration, decision making and operation of the controlled environment. The environment for the prototype is simplified, yet meaningful because the target is not fixed for each demonstration, but instead is set according to the real-time sensor data. The system is designed to take decisions regarding availability, authorization and condition of the parking slots. Not only can general assumptions be tested, but so can repeatable experiments be undertaken to test the performance of the project. The allocation/display, gate/guidance response should be the same as with each authorised RFID card. The ability to detect a parking space as it changes from empty to occupied and occupied to empty should be robust. The robot should successfully navigate to Slot 1 and 2 and 3 from the same starting point and stop when obstacles are positioned at a number of different distances. Other cases, such as full parking, invalid RFID, expired reservation, sensor failure etc should also be tested. The main aims of the project are: Manual control between different operations is not necessary during the whole operation process of RFID verification, slot allocation, reservation, gate opening and robot guidance. Practical decision making: Do not allow duplicate assignments of slot, Do not allow to park if no appropriate slot, Ignore incorrect or uncertain sensor reading as fault (not slot). Measurable performance: Test the system on a well-defined performance measure, e.g., rate of success in navigation, distance from which obstacle is detected, response time to the allocation, accuracy of detecting the slot in repeated tests.

Societal Implications

Parking issues are pervasive in shopping malls, office buildings, apartment complexes, universities and hospitals, etc. It can impact on both drivers and passengers, pedestrians, the security of the area, delivery workers, and traffic around the area. This can eliminate the need for any unnecessary searching, minimize unnecessary movements in parking lots and facilitate the overall parking procedure by establishing a parking destination for every authorized vehicle. This can also also help to relax congestion on the entrance in peak hours, and reduce the time that security personnel need to spend manually diverting vehicles. A guidance robot would also assist the visitors, who are unfamiliar with the parking system, and provide them direction. The low cost design of this prototype opens the possibility for the smaller organisations to consider automation without the expensive

camera based monitoring system or autonomous industrial vehicle. However, there is no correlation between the price of a product and access to it. If there aren't any spaces available, for instance, visitors without registered RFID cards might not be able to enter. If implemented in a real world environment, other means of access would be required, such as temporary visitor's cards, an intercom system, staff approval or another means to verify manually. Special parking areas like staff parking, charging stations, emergency parking and disability parking also need to be managed separately. There should not be a simple nearest slot allocation process in place that allocates these spaces to users that are not eligible to use it. Parking staff can be automated, but can't be replaced. Manual gate opening/closing and traffic direction may be decreased, sensor maintenance, battery management, system monitoring, record-keeping and user support may be increased. Training before deployment and staff being able to take manual control at all times should be provided in case of unexpected situations. The system also depends on the information shared by the sensors, so if the information is wrong on one sensor, then it could affect multiple users as they would be allocated to the wrong sensor. Therefore, the system should be able to detect faulty elements, recognize impacted parking areas, and run in a reduced but comprehensible manner rather than obscure or make unsafe assumptions. Human interaction should not be eliminated from the decision-making process, but a reliable automated system should assist humans in the decision-making process.

Health Implications

The biggest health benefit of this system is that it may help to minimize an unneeded movement of cars on and off parking lots. Drivers are given their parking spot prior to entering the parking lot, which reduces idling time and searching for a parking spot. If the parking area is enclosed or partially enclosed, minimizing unnecessary traffic can help reduce the exposure to vehicle exhaust in the parking area, particularly at peak times when many vehicles are entering and exiting the area simultaneously. Giving clear instructions can also lessen driver stress and mental load, since users don't have to keep moving through a number of parking rows, checking out pedestrians and other cars.
Several health and safety issues arise along with the prototype, however, regarding the electronic components, the batteries, the moving mechanisms and the warning systems. Leakage, short circuiting or overheating are threats that can occur with the damaged or wrongly charged rechargeable battery. Open or exposed electrical connections are dangerous and if it is a continuous alarm, it may cause discomfort to the user. These hazards can be minimised by ensuring that the wiring is properly insulated, the battery is in a robust holder, the charging process is suitable and the power switch is easily accessible, and short warning signals are at an acceptable sound level.

Physical state of the robot and its environment also should be taken into account. Around the components, dust and debris can build up, which can cause inaccuracy of the sensor measurement and influence the reliability of the sensor operation. Regular cleaning and maintenance is necessary to keep devices performing in a consistent manner and exclude the need for the user to tamper with the equipment when it is not performing properly.

The system can be used to minimize unnecessary searching and vehicle movement, but should not be used as a substitute for proper environmental safety systems. Emissions are also influenced by other factors

including the quality of the ventilation, the number of vehicles and vehicle condition and cannot be guaranteed by parking guidance. A proper ventilation system, as well as separate fire and air protection systems, would still be required for real parking facilities, according to the building standards.So the project should make a realistic claim: it can help to reduce unnecessary movement of vehicles and idling times, but will not be able to eliminate the health impacts caused by vehicle emissions.

Safety Implications

The most important factor in this project is safety, since this system will control the automatic entrance barrier, be used by a moving robot near vehicles and pedestrians, and also be used by the robot to perform safety-related tasks. Each automated action requires several components such as sensors, software decision making, communication links, and mechanical components. All of these factors can malfunction or deliver wrong data. Hence, if uncertain data is received by the system, it must halt the corresponding operation, avoid unsafe operations and ask for human help.
The robot should be controlled to run at a steady speed so that it can stop safely within the limits of the distance available in the prototype environment. It should be on a well-defined route to minimise the risk of unexpected actions. Some items, particularly narrow and soft surfaces, might emit lower readings at the ultrasonic obstacle sensor and so the distance and angle must be tested. Likewise, slot sensors should need a number of steady readings to update the slot's status so that the status may not fluctuate from one read to another and cause the sensor to allocate a slot that is not actually available.
If the sensor value is not in the range the sensor is expected to see then that sensor value should be considered as a Fault condition instead of a valid space to park. This will ensure that no slot is allocated which may not be available. The entrance barrier needs also to be considered. Should stay open until it is confirmed that a vehicle has passed; this can be done using a passage sensor, a verified delay period or manual confirmation. The system should have a manual override if the servo motor, sensor or control program should operate in an unexpected manner.

The final demonstration should only be a miniature prototype demonstration and should not be given as a safety certified system for real vehicles. A complete scale deployment requires further testing, industrial parts and a professional safety validation.The main safety requirements are:

- **Fail-safe stopping:** The robot must stop its motors when an obstacle is detected within the tested safety range, when communication data is incomplete, or when an invalid movement command is received.
- **Safe gate control:** The barrier should remain open until safe passage is confirmed and should include a manual opening option if automatic control fails.
- **Emergency control:** An accessible emergency-stop button should be included to immediately disable robot movement without relying on RFID, wireless communication, or the main software sequence.
- **Protected construction:** Batteries and circuit boards should be securely mounted, exposed terminals should be covered, sharp wires should be removed, and users should receive a warning signal before the robot begins moving.

Legal Implications

For RFID access control, privacy and data management issues are a concern. Information on the tag can contain the date of scanning the tag, the time-series of access authorizations for the tag, and information about the parking space assigned to the tag. In a real-life situation, information required for parking management should be gathered and the rationale for data gathering, who will have access to the information and how long it will be retained should be explained to the operator. Access to records must not be public and records must not be held unless there is a good reason for holding them. Records should be removed in accordance with policies and regulations of the organisation and administrative access should be limited.

The ownership of the failures of the system should also be specified. If the guidance is incorrect some users may be directed to an occupied parking space, if the barrier fails, a car may be damaged and if there is an authorization error, a legitimate user may be denied access to the parking area. Proper testing records, maintenance records, warning signs and manual override features may indicate good operations, but without any of these features it is not enough to remove the organizational responsibilities of the system's operation. It would have to be formally risk assessed and meet requirements for private data, electrical safety, wireless communications, fire protection, accessibility, workplace safety and building standards in a commercial version. The definition of the role of the RFID should be precise too. An RFID card can only determine whether a recognized tag has been presented to the system; it is not to say who has used it or whether it would belong to the owner of a vehicle. Hence, it is not advisable to promote the system as a full-fledged security solution.
It is important to make it clear in the project documentation and presentation that the system is a low speed educational prototype to be demonstrated. If the system is to be deployed with actual vehicles, then the system must be made more robust with industrial equipment, added safety features, properly engineered and tested, and approved by appropriate agencies.

Cultural Implications

The level of compliance with the instructions given by the robot to drive to the parking spot may differ depending on the age and experience of the driver, and/or on how used to automated systems they are. For some, it may be simpler and more efficient to receive directions from a robot than from a human employee, and some may be angry to have directions given automatically. The user must be able to trust the robot by following predictable motions, seeing what's on the screen about where the robot is going, seeing the pathway markings, and being able to ask for human assistance when needed. The system should be simple and understandable to the users. Use short and familiar messages on the LCD display (e.g., "Slot 2", "Parking Full", "Invalid Card", "Please Wait") and avoid complex technical instructions. At sites that have a variety of users, vital data might have to be given in different languages. The use of only red and green colours may make this difficult for users with colour-blindness or users who may have a different understanding of these colours than other cultures. Therefore, the slot numbers, written message, LED positions and sound indicators need to coordinate and be clear about the system's status.

Local traffic patterns, parking etc. should also be considered when deployed. Some users might follow too close to the robot, try to pass the robot or not follow the robot's stopping signals. To minimize confusion,

the parking area should have clearly marked signs, a controlled one-way path and directions as to how the parking system should be used.

A small pilot test should be performed to see how people use the system, to be able to identify common misunderstandings, gather feedback and enhance the interface and navigation process before implementing the system on a larger scale. Evaluating cultural appropriateness should be done as part of their existing use rather than design.

Conclusion

Unlike the conventional parking availability indicator, the Autonomous Parking Guide Robot with Smart Slot Allocation integrates various functions into one system. The prototype integrates the following functions: RFID-based authorization, real-time parking slot monitoring, reservation management, parking slot allocation, automatic gate, wireless communication, robot navigation, obstacle detection and exception handling into a full workflow.
The advantage that this has to the user is an easier parking experience as people need to spend less time searching for parking and will be guided to the correct area to park, as vehicles move more smoothly and will not idle unnecessarily. The project is also educational because it demonstrates the potential of cooperation and interaction between the various embedded systems and robotics elements for decision making, exchange of information and effective automation.

The project can be realized in this 1-month development time as it is restricted to a small-scale prototype parking environment with only 3 parking spaces and a limited number of parking patterns and a fixed navigation path. Success of the system should be judged on reliability not features. A high performance of the navigation, proper fault handling, safe motion, manual control abilities and well-defined limits will improve the effectiveness and convince of the demonstration. The system should be shown as a small-scale model to illustrate the feasibility of automated parking instead of offering a complete commercial parking system. This is not a realistic implementation, as that would necessitate more extensive testing, industrial-grade hardware, further safety features and professional certification.
The project can be considered successful when it achieves the following goals:

- **Functional success:** A valid RFID scan should consistently result in correct slot reservation, display information, barrier operation, and robot guidance while preventing duplicate allocation and unauthorized entry when parking is unavailable.
- **Safe success:** Obstacle detection, emergency stopping, secure barrier operation, protected wiring, and clear fault messages should continue to function even when the normal operating sequence is interrupted.
- **Responsible success:** The system should protect access information, support different categories of users, provide human assistance when required, and clearly communicate that real vehicle deployment requires professional evaluation and certification.